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Header height correction system and method

Abstract

An agricultural machine includes a main frame, a rockshaft pivotably coupled to the main frame, a lift actuator couple between the rock shaft and the main frame, a lift arm pivotably coupled to the main frame below the rockshaft, a lift strap coupled between the rockshaft and the lift arm, and a lift actuator coupled between the main frame and the lift strap to pull the lift arm upwards. The agricultural machine includes header, which is supported by the lift arm and the float actuator above the ground at a desired harvesting height. During harvesting, the lift arm may encounter uneven ground, causing the header to move away from the desired harvesting height. By measuring float actuator pressure or header position and by changing the pressure in the float actuator or lift actuator in response, the header can be rapidly returned to the desired harvesting height.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure relates to controlling the height of a header for an agricultural machine and, in particular, to rapidly adjusting the height of the header.

BACKGROUND OF THE DISCLOSURE

(2) Commercial mowers have varying types of cutting, collecting, and conditioning components. Some mowing machines may only provide a cutting function and are without any crop processing or collecting capability while other machines serve all three functions. Some agricultural machines include a frame, which may be referred to a traction assembly, which is capable of supporting and moving different headers for different cutting, collecting, and conditioning operations.
(3) In use, a header may be located at various heights throughout the machine's operation. For example, the header may raise in response to contacting a bump in the field; it may lower in response to entering a ditch in the field; or it may alternate between repeated up and down movement in response to uneven terrain. Moreover, the machine may move from the field to the road (or visa-versa) or may make a headland turn at the end of the field before proceeding back in the opposite direction. In conventional machines, the header may be located at an undesirable height during certain instances described above. In other instances, the headers of conventional machines may be moved to a desired position too slowly such that harvested crop yield is reduced.
(4) Therefore, what is needed is a system and method for rapidly adjusting the height of a header in response to sensed or stored criteria associated with the terrain or the agricultural machine.
SUMMARY
(5) In an illustrative embodiment, an agricultural machine for rapidly moving a header to a harvesting position comprises: a main frame supported above the ground by ground engaging mechanisms; a lift arm pivotably coupled to the main frame; the header which is configured to harvest crop, wherein the header is coupled to and supported above the ground by the lift arm; a lift actuator configured to extend and contract to pivot the lift arm and the header relative to the main frame; a float actuator pivotably coupled to the lift arm and the main frame to support the header above the ground; at least one valve fluidly coupled to the float actuator and configured to move between open and closed positions to change pressure in the float actuator; a controller configured to send electrical signals to the at least one valve to execute a rapid height correction algorithm, in which the controller adjusts at least one of the float actuator and the lift actuator causing a change in height of the header relative to the ground, in response to receiving correction data associated with at least one of a current height of the header and a current pressure of the float actuator.
(6) In some embodiments, the agricultural machine further comprises a position sensor operatively coupled to the controller and configured to measure a current height of the header relative to the main frame or relative to the ground; wherein the controller is configured to receive correction data from the position sensor associated with the current height of the header; wherein the controller is configured to compare the current height of the header to the harvesting position of the header, which is a predetermined desired height for the header during a harvesting operation; and wherein the controller is configured to execute the rapid height correction algorithm in response to determining that the current height of the header is beyond a threshold difference from the harvesting position of the header.
(7) In some embodiments, in response to determining that the current height of the header is above the harvesting position of the header, the controller sends electrical signals to the least one valve causing the float actuator to extend.
(8) In some embodiments, agricultural machine further comprises a pressure sensor operatively coupled to the controller and configured to determine the current pressure of the float actuator; wherein the controller is configured to receive correction data from the pressure sensor indicative of the current pressure of the float actuator; and wherein the controller is configured to execute the rapid height correction algorithm in response to receiving the correction data from the pressure sensor.
(9) In some embodiments, if the correction data received from the pressure sensor is associated with a current height of the header above the harvesting position, then to execute the rapid height correction algorithm the controller sends electrical signals to the least one valve causing the float actuator to extend.

(10) In some embodiments, to cause the float actuator to extend, the controller is configured to send a first electrical signal to a first valve causing the first valve to move toward the open position and a second electrical signal to a second valve causing the second valve to move toward the open position.

(11) In some embodiments, the correction data includes a signal from the pressure sensor indicative of a decrease in current pressure in the float actuator; and in response to the signal from the pressure sensor indicative of a decrease in current pressure in the float actuator, the controller is configured to send electrical signals to the least one valve causing a further decrease in the current pressure of the float actuator. In some embodiments, the correction data includes a signal from the pressure sensor indicative of an increase in the current pressure of the float actuator; and in response to the signal from the pressure sensor indicative of an increase in the current pressure of the float actuator, the controller is configured to send electrical signals to the least one valve causing a further increase in the current pressure of the float actuator.

(12) In some embodiments, the controller is configured to analyze terrain data including ground contours to determine whether to send electrical signals to the least one valve causing a further increase in the current pressure of the float actuator. In some embodiments, if the correction data received from the pressure sensor is associated with a current height of the header relative below the harvesting position, then to execute the rapid height correction algorithm the controller sends electrical signals to the at least one valve causing the lift actuator to contract.

(13) In some embodiments, if the correction data received from the pressure sensor is associated with a current height of the header below the harvesting position, then to execute the rapid height correction algorithm the controller sends electrical signals to the at least one valve causing the float actuator to contract.

(14) In some embodiments, subsequent to execution of the rapid height correction algorithm, the controller is configured to compare a predetermined pressure of the float actuator to the current pressure of the float actuator; and the controller is configured to send additional signals to the at least one valve to adjust the current pressure of the float actuator to match the predetermine pressure of the float actuator.

(15) In some embodiments, the correction data comprises global positioning data associated with the agricultural machine performing a 180 degree turn and moving between a field and a road. In some embodiments, the correction data comprises a predetermined operating sequence of the agricultural machine associated with at least one of the agricultural machine performing a 180 degree turn and the agricultural machine moving between a field and a road.

(16) In another illustrative embodiment, an agricultural machine for rapidly moving a header to a harvesting position comprises: a main frame supported above the ground by ground engaging mechanisms; a rockshaft pivotably coupled to the main frame for rotation about a first axis; a lift actuator coupled at a first end to the main frame and at a second end to the rockshaft to facilitate pivoting motion of the rockshaft relative to the main frame; a lift arm pivotably coupled to the main frame for rotation about a second axis parallel to the first axis; a header configured to harvest crop, the header being coupled to and supported above the ground by the lift arm; a float actuator pivotably coupled to the lift arm for rotation about a third axis, the third axis being parallel to the first axis, wherein the float actuator provides an upward force to the lift arm, thereby cooperating with the lift arm to support the header above the ground; at least one valve fluidly coupled to the float actuator and configured to move between open and closed positions to change the pressure in the float actuator; and a controller configured to send electrical signals to the at least one valve to execute a rapid height correction algorithm, which extends or contracts the float actuator causing a change in height of the header relative to the ground, in response to receiving correction data associated with at least one of a current height of the header and a current pressure of the float actuator; wherein contraction of the float actuator is configured to occur independent of pivoting motion of the rockshaft relative to the main frame.

(17) In another illustrative, a method of operating an agricultural machine to rapidly return a header of the agricultural machine to a harvesting position comprises: receiving, from a sensor of the agricultural machine, correction data associated with a current position of the header; comparing the correction data with the harvesting position of the header, which is a predetermined value associated with a desired height of the header during a harvesting operation; and adjusting a float actuator, which supports the header above the ground, to move the header to the harvesting position, in response to the comparison between the correction data and the harvesting position, wherein the float actuator is coupled between a lift arm that assists in supporting the header above the ground and a main frame of the agricultural machine to which the lift arm is pivotably coupled.
(18) In some embodiments, the method further comprises, prior to the receiving step, measuring a current pressure of the float actuator via the sensor to obtain the correction data. In some embodiments, the method further comprises, prior to the receiving step, measuring the height of the header relative to the ground via the sensor to obtain the correction data. In some embodiments, the method further comprises, prior to the receiving step, measuring the height of the header relative to the main frame of the agricultural machine via the sensor to obtain the correction data.
(19) In some embodiments, the adjusting step includes: sending electrical signals to a first valve and a second valve, which are fluidly coupled to the float actuator to cause movement of the first valve and the second valve between open and closed positions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The above-mentioned aspects of the present disclosure and the manner of obtaining them will become more apparent and the disclosure itself will be better understood by reference to the following description of the embodiments of the disclosure, taken in conjunction with the accompanying drawings, wherein:
(2) FIG. **1** is a side front perspective view of an agricultural machine including a mower implement positioned at a front end of the agricultural machine;
(3) FIG. **2** is a side front perspective view of the agricultural machine including a mower-conditioner implement positioned at a front end of the agricultural machine;
(4) FIG. **3** is a side front perspective view of a traction assembly of the agricultural machine of FIGS. **1** and **2**;
(5) FIG. **4** is a schematic or diagrammatic view of a hydraulic system for operating the traction assembly of FIG. **3**;
(6) FIG. **5** is a diagrammatic view of a control system for the agricultural machine of FIGS. **1** and **2** having a controller, a height sensor, and a plurality of valves that are controlled by the controller;
(7) FIG. **6** is a flow chart showing a process of identifying and correcting header height for a header coupled to the traction assembly of FIG. **3**, wherein the header is another name for the implements of FIGS. **1** and **2**;
(8) FIG. **7** is a flow chart showing in detail the process of detecting a header height correction scenario; and
(9) FIG. **8**A is a flow chart showing in detail the process of detecting a header height correction scenario and correcting header height;
(10) FIG. **8**B is a flow chart showing a continuation of the process of detecting a header height correction scenario and correcting header height.
(11) Corresponding reference numerals are used to indicate corresponding parts throughout the several views.
DETAILED DESCRIPTION
(12) The embodiments of the present disclosure described below are not intended to be exhaustive

or to limit the disclosure to the precise forms in the following detailed description. Rather, the embodiments are chosen and described so that others skilled in the art may appreciate and understand the principles and practices of the present disclosure.

(13) Referring to FIG. **1**, an agricultural machine **10** is shown and is operable to harvest crop in a field. The agricultural machine **10** includes a main frame **12** supported on right and left front wheels **14**R and **14**L, respectively, and on right and left caster mounted rear wheels, of which only a left rear wheel **16**L is shown. In some embodiments, other wheel configurations (and/or tracks) may be included in the agricultural machine **10**. The agricultural machine **10** also includes a cab **18** carried on a forward portion of the main frame **12**. Operator controls (e.g. a user interface) are provided in the cab **18** for operation of the agricultural machine **10**. In the illustrative embodiment, the agricultural machine **10** includes a housing **20** positioned reward of the cab **18** and a power source such as an internal combustion engine positioned in the housing **20**. The agricultural machine **10** also includes a traction assembly **36** coupled to the forward portion of the main frame **12**. As shown in FIG. **1**, the agricultural machine **10** includes a header **22**, which may be referred to as a draper or mower implement, coupled to and supported above the ground by the traction assembly **36**.

(14) The header **22** includes a laterally extending frame **24** that extends perpendicular to a fore-and-aft harvesting direction of travel "V" of the agricultural machine **10**. The header **22** further includes an elongate reciprocating knife **26** that extends across a width of the header **22**. This reciprocating knife **26** is disposed immediately in front of a harvested crop conveyor assembly. In the illustrative embodiment, the harvested crop conveyor assembly includes a right side conveyor **28** that carries harvested crop from the right side of the header **22** to a central region of the header **22**, a left side conveyor **30** that carries crop from the left side of the header **22** to the central region of the header **22**, and a central conveyor **32** that receives crop from the left side conveyor **30** and the right side conveyor **28**, and conveys the harvested crop rearward.

(15) In this disclosure, the implement (i.e. header) of the agricultural machine **10** could take many configurations in addition to that described with regard to FIG. **1**. For example, as shown in FIG. **2**, the implement of the agricultural machine **10** may be embodied as a mower-conditioner implement, also referred to as a header **34**. The header **34** may include an impeller (e.g., an elongated roll) with a plurality of tines used to scrape harvested crop from cutting blades to shorten drying time of the crop. In other embodiments, the header **34** may include a roll conditioner (e.g., a pair of cylindrical rolls) used to crimp crop stems to shorten drying time and mitigate exposure to environmental elements. In any event, as shown in FIG. **2**, the header **34** includes a laterally extending frame **38** that extends perpendicular to the fore-and-aft harvesting direction of travel "V" of the agricultural machine **10**.

(16) FIG. **3** is a perspective view of the traction assembly **36**. The traction assembly **36** includes a rockshaft **40** coupled to the forward portion of the main frame **12**. The rockshaft **40** illustratively includes a cylindrical bar and a plurality of plates coupled to the cylindrical bar for pivoting motion therewith. The traction assembly **36** further includes a header lift actuator **42**, which is coupled to the rockshaft **40** to cause pivoting motion thereof relative to the main frame **12**. For example, when the lift actuator **42** contracts, the rockshaft **40** pivots upward relative to the main frame **12**, and when the lift actuator **42** extends, the rockshaft **40** pivots downward relative to the main frame **12**. As suggested by FIG. **3**, the rockshaft **40** pivots about an axis **46**, which extends through the cylindrical bar, relative to the main frame **12**. In the illustrative embodiment, the lift actuator **42** is a hydraulic actuator; however, this disclosure contemplates electric and other actuators as well.

(17) Referring still to FIG. **3**, the traction assembly **36** further includes a first lift arm **48** and a second lift arm **50** each positioned below and spaced apart form the rockshaft **40**. In the illustrative embodiment, the lift arms **48**, **50** are identical to each other. The lift arms **48**, **50** are pivotably coupled to the main frame **12** and configured to pivot about an axis **52** relative to the main frame **12**. In the illustrative embodiment, the axis **52** is parallel to the axis **46**. A header (e.g., header **22** or

34) may be coupled and supported by the lift arms **48**, **50** to facilitate a mowing operation. The traction assembly **36** further includes a first lift strap **54** and a second lift strap **56**, each being pivotably coupled at a first end to a plate of the rockshaft **40** and pivotably coupled at a second end to lift arms **48**, **50**, respectively. In the illustrative embodiment, the lift straps **54**, **56** are identical to each other. The first end of each lift strap **54**, **56** includes at least one slot **60** that facilitates vertical sliding movement of the lift straps **54**, **56** relative to the rock shaft **40**, which in turn facilitates additional pivoting movement of the lift arms **48**, **50** relative to the main frame **12**. The vertical sliding movement of the lift straps **54**, **56** facilitates movement of the lift arms **48**, **50** and header **22** without movement of the rockshaft **40**. While the header **22** is referred to above, it should appreciated the disclosure above and otherwise herein is applicable to headers **22**, **34** and other headers suitable for harvesting operations described herein.
(18) The traction assembly **36** further includes first and second float actuators **64**, **66**, which are pivotably coupled at first ends to the main frame **12** and at second ends to middle portions of the first and a second lift straps **54**, **56**, respectively. In the illustrative embodiment, the middle portion of each lift strap **54**, **56** is positioned between the first and second end thereof. The float actuators **64**, **66** contract with upward movement of the lift arms **48**, **50** and extend with downward movement of the lift arms **48**, **50**.
(19) As shown in FIG. **4**, the agricultural machine **10** includes a hydraulic system **70**, which includes the float actuators **64**, **66**. As suggested by FIG. **3**, the weight of the header **22** on the lift arms **48**, **50** provides a downward force, urging the lift arms **48**, **50** to pivot downward relative to the main frame **12** about the axis **52**. The float actuators **64**, **66** are pressurized to counteract the downward force on the lift arms **48**, **50**. As the agricultural machine **10** travels, the lift arms **48**, **50** may encounter uneven terrain (e.g., a bump) causing the lift arms **48**, **50** to rapidly pivot (e.g., upward) relative to the main frame **12**, which may include rapidly sliding of the lift straps **54**, **56** relative to the rock shaft **40** (e.g., upward), which in turn causes a change in the arrangement of the float actuators **64**, **66** (e.g., contraction). Contraction or extension of the float actuators **64**, **66** causes a change in the amount of pressurized fluid in accumulators **68**, **72** of the hydraulic system **70**.
(20) Subsequent to the machine's initial response to encountering the uneven terrain, the pressure in the hydraulic system **70** equalizes such that the lift arms **48**, **50** slowly return to the height at which the lift arms **48**, **50** were positioned prior to the encountering the uneven terrain, which is a predetermined height or a height set by the operator, referred to as the harvesting position. It is desired to rapidly return the header **22** to the harvesting position. As used herein, the slowly and rapidly are terms used relative to each other to describe the improved rate of return to the harvesting position (leading to improved harvesting yield), which is associated with the system and methods disclosed herein versus that of conventional agricultural machines. For example, in convention machines, after contacting a bump, approximately five seconds may pass before a header returns to the harvesting position, which in some embodiments equates to approximately 10-15 square meters of unharvested crop.
(21) It has been identified that other events such as a header dropping into a ditch, a header bouncing up and down, an agricultural machine passing from road to field or visa-versa, and an agricultural machine taking a headland turn may each cause a header to move between a harvesting position and another position at a rate that is slower than desired. The processes described herein may be used to manipulate the hydraulic system **70**, such that the hydraulic system **70** moves the lift arms **48**, **50** (and thereby the header **22**) to the harvesting position more rapidly.
(22) As shown in FIG. **4**, the hydraulic system **70** includes the float actuators **64**, **66** and the float accumulators **68**, **72**. It should be appreciated that the float actuators **64**, **66** may be pressurized independently from one another based on the hydraulic arrangement shown in FIG. **4** and described below. The hydraulic system **70** also includes several valves illustratively embodied as electrically-controlled two-position solenoid valves. In other embodiments, the valves may be variable. In any

event, the hydraulic system **70** includes a center valve **74**, a right float pressure valve **76**, a left float pressure valve, **78**, a header raise valve **80**, and a header lower valve **82**. The center valve **74** is biased to remain open when not electrically activated, and the other valves **76**, **78**, **80**, **82** are biased to remain closed when not electrically activated. The components of the hydraulic system **70** may be referred to as fluidly coupled to one another in a hydraulic circuit as shown in FIG. **4**. Valves that are being electrically activated or powered may be referred to as "on" while valves that are not being electrically activated or powered may be referred to as "off."

(23) As suggested by FIG. **4**, when the center valve **74** is in the closed position (i.e. "on") and the right float pressure valve **76** is in the open position (i.e. "on"), the pressure in the float actuator **64** increases. In contrast, when the center valve **74** is in the open position (i.e. "off") and the right float pressure **76** is in the open position (i.e. "on"), the pressure in the float actuator **64** decreases. When the center valve **74** is in the closed position (i.e. "on") and the left float pressure valve **78** is in the open position ("on"), the pressure in the float actuator **66** increases. And in contrast, when the center valve **74** is in the open position (i.e. "off") and the right float pressure valve **78** is in the open position ("on"), the pressure in the float actuator **66** decreases. Increase and decrease of pressure in one float actuator (e.g., **64**) may occur independently of that in the other float actuator (e.g., **66**). In contrast, raising and lowering of the traction assembly **36** via pivoting motion of the rockshaft **40** occurs in response to extension or contraction of a single lift actuator **42**. To raise the traction assembly **36** via the lift actuator **42**, the center valve **74** must be in the closed position (i.e. "on") and the header raise valve **80** must be in the open position (i.e. "on") such that the lift actuator **42** contracts. To lower the traction assembly **36** via the lift actuator **42**, the header lower valve **82** must be in the open position (i.e. "on") such that the lift actuator **42** extends. These principles of operation are used in connection with the methods described below to rapidly move the header **22** to the harvesting position, as described below.

(24) Referring now to FIG. **5**, a diagrammatic view of a control system **84** of the agricultural machine **10** is shown. The control system **84** includes a controller **86** having at least one memory and at least one processor configured to execute instructions (i.e., algorithmic steps; see FIGS. **6-8**, for example) that are stored on the at least one memory. The controller **86** may be a single controller or a plurality of controllers operatively coupled to one another. The controller **86** may be housed by the agricultural machine **10** or positioned remotely, away from the agricultural machine **10**. The controller **86** may be hardwired or connected wirelessly to other components of the agricultural machine **10** via Wi-Fi, Bluetooth, or other known means of wireless communication.

(25) As suggested by FIG. **5**, the controller **86** is operatively coupled to the plurality of electrically-controlled valves of the hydraulic system **70**. The controller **86** is configured to send signals to the valves **74**, **76**, **78**, **80**, **82** causing the valves to move between open and closed positions. In the illustrative embodiment, the controller **86** is also operatively coupled to at least one lift arm sensor **88**, which is illustrative positioned on a forward end of one or both lift arms **48**, **50**, as shown in FIG. **3**. The at least one lift arm sensor **88** may also be referred to as a position sensor **88**. In other embodiments, the position sensor **88** may be configured to measure angular position, such as the angle of the lift arms **48**, **50** relative to a portion of the main frame **12**.

(26) In any event, the controller **86** is configured to receive signals from the at least one position sensor **88** indicative of the height of one or both lift arms **48**, **50** relative to the ground, relative to the main frame **12**, or relative to another component of the agricultural machine **10** having a known height. If a height of the main frame **12** relative to the ground is known, for example, a position of one or both lift arms **48**, **50** relative to the main frame **12** may be used as a proxy for the height of one or both lift arms **48**, **50** relative to the ground. Therefore, in the illustrative embodiment, the at least one position sensor **88** may be any sensor(s) suitable to measure position or rotation of the lift arms **48**, **50** directly or indirectly relative to the ground or relative to another component of the agricultural machine **10**.

(27) In some embodiments, the controller **86** is operatively coupled to first and second pressure

sensors **92**, **94** that are configured to measure the pressure in the float actuators **64**, **66**, respectively. In other embodiments, the pressure sensors **92**, **94** may be positioned in other locations of the hydraulic system **70** so long as the sensors are suitable to measure the in-line pressure of the hydraulic circuit corresponding to the pressure of the float actuators **64**, **66**. The pressure sensors **92**, **94** are configured to send signals to the controller **86** indicative of the pressure in the float actuators **64**, **66**.

(28) The controller **86** may be operatively coupled to a user interface **90** (e.g., the user interface positioned in the operator cab **18**) and configured to receive input data from the user via the user interface **90**. Moreover, the controller **86** may output data to the user interface **90** such as data indicating the height of the lift arms **48**, **50** or the pressure of the float actuators **64**, **66**.

(29) As shown in FIG. **6**, a method **600** of correcting header height includes steps **602**, **604**, and **606**. At step **602**, the controller **86** determines that a height header correction scenario (i.e., a scenario prompting the need for rapid change in header height) has been detected. At step **604**, the controller **86** executes a header height correction algorithm by sending signals to at least one of the electrically operated valves **72**, **74**, **76**, **80**, **82** to cause the valves to open or close; this may be referred to as turning "on" the valves. At step **606**, the controller **86** stops sending electrical signals to the valves **72**, **74**, **76**, **80**, **82** which it was previously sending signals to in order to stop actuation of the lift actuator **42** and/or the float actuators **64**, **66**; this may be referred to as turning "off" the valves. At step **608** the controller **86** determines whether the pressure of a float actuator **64**, **66** (e.g., as measured by pressure sensors **92**, **94**) is equal to a desired pressure value, which may be predetermined or set by an operator of the agricultural machine **10**. If the measured pressure is not equal to the desired pressure value, then at a step **610** the controller **86** is configured to send signals to the valves **72**, **74**, **76**, **80**, **82** causing the valves to open or close to adjust the pressure to the desired pressure value.

(30) Referring now to FIG. **7**, a more detailed disclosure of the process **602** of detecting a header height correction scenario is shown. It should be appreciated that steps of detecting a header height correction scenario may be performed in any order. Steps **612**, **614**, and **616** include determining whether the header **22** and/or lift arms **48**, **50** have encountered uneven terrain urging the header **22** upward or downward. In the illustrative embodiment, the agricultural machine **10** is able to determine when the header **22** encounters uneven terrain such as a bump, a ditch, or repeatedly alternating uneven terrain. The determination is made by the controller **86** based on receipt and analysis of correction data, which includes at least one of (i) a current position of the header **22** relative to the ground or relative to the main frame **12** and (ii) a current pressure value representative of the pressure of a float actuator (i.e., **64** or **66**). While FIG. **8**, for example, describes the "position threshold," it should be appreciated that the valves and actuators described herein may also be adjusted by the controller **86** in response to measured pressure values received by the controller **86**. For example, in some embodiments, pressure in the hydraulic system **70** is used as a proxy or value associated with the height of the header **22**. In other embodiments, the position sensor **88** is used to measure the height of the header **22** independent of a pressure measurement.

(31) In the illustrative embodiment shown in FIG. **8**, the controller **86** compares the measured height of the header **22** to the harvesting position of the header **22**, which is a desired height for the header during a harvesting operation. The controller **86** executes the rapid height correction algorithm if the measured height of the header **22**, is above or below a threshold difference from the desired height. For example, at step **624**, the controller **86** determines based on correction data received from the position sensor **88** whether the header **22** is above the position threshold. If so, then at a step **626**, the controller **86** sends electrical signals to turn off the center valve **74** and turn on a float valve **76**, **78** to reduce the pressure in the float valve **76**, **78**, which facilitates rapid movement of the header **22** to the harvesting position. At steps **628** and **630**, if the controller **86** determines based on the position sensor **88** that the header **22** is in the harvesting position, (i.e.

below a lower position threshold), then in response, the controller **86** turns off the center valve **74** (or ensures that the valve is off) and turns off the float valve **76**, **78**. In other embodiments, the controller **86** determines based on correction data received from the pressure sensors **92**, **94** whether a decrease in pressure at a float actuator **64**, **66** beyond a threshold pressure has occurred. If so, then the controller **86** sends electrical signals to turn off the center valve **74** and turn on a float valve **76**, **78** to reduce the pressure in the float valve **76**, **78**, which facilitates rapid movement of the header **22** to the harvesting position. Thereafter, the controller **86** sends electrical signals to turn off the center valve **74** and the float valves **76**, **78**, as described above.

(32) Referring still to FIG. **8**, at step **632**, the controller **86** determines based on correction data received from the position sensor **88** whether the header **22** is below the position threshold. If so, then at a step **634**, the controller **86** determines whether terrain data is stored in the memory. If so, then the controller **86** determines whether the header **22** is passing through a ditch or advancing along a continuing downslope. If the header **22** is advancing along a continuous downslope then no action is taken; however, if the header **22** is below the position threshold because it has entered a ditch, then at step **636**, the controller **86** sends electrical signals to turn on the center valve **74** and turn on the header raise valve **80** to lift the header **22** to avoid contact with the upward-sloping side of the ditch. At step **636**, it is also contemplated that the controller **86** may send electrical signals to turn on the center valve **74** and turn on the float valves **76**, **78** thereby raising the float pressure in the float actuators **64**, **66** to raise the header **22**. Thereafter, as shown at step **638**, the controller **86** sends electrical signals to turn off the center valve **74**, the float valves **76**, **78**, and the header raise valve **80**.

(33) Referring again to FIG. **7**, the process **602** of detecting a header height correction scenario also include steps **618**, **620**, and **622**, which include determining whether the agricultural machine **10** is moving from the field to the road (i.e. step **618**), moving from the road to the field (i.e. step **620**), or taking a headland turn (i.e. step **622**), which may also be described as a 180 degree turn often occurring at the boundary of a field or the end of a planted portion thereof. In the illustrative embodiment, the agricultural machine **10** follows a known operating sequence indicative of each step **618**, **620**, **622** described above. For example, when the agricultural machine is making a headland turn, the header **22** is raised and the ground engaging mechanisms (e.g. wheels **14**, **16**) are turned; likewise when the agricultural machine **10** moves from the field to the road, the header **22** is raised; and likewise when the agricultural machine moves from the road to the field, the header **22** is lowered. The controller **96** compares the current operating sequence of the agricultural machine **10** to operating sequences stored in the memory to determine whether the agricultural machine **10** is moving from the field to the road (i.e. step **618**), moving from the road to the field (i.e. step **620**), or taking a headland turn (i.e. step **622**). In some embodiments, based on global positioning (GPS) data provided to the controller **86**, the controller **86** is determines whether the agricultural machine **10** is moving from the field to the road (i.e. step **618**), moving from the road to the field (i.e. step **620**), or taking a headland turn (i.e. step **622**). In some embodiments, based on operator input to the user interface **90** received by the controller **86**, the controller **86** determines whether the agricultural machine **10** is moving from the field to the road (i.e. step **618**), moving from the road to the field (i.e. step **620**), or taking a headland turn (i.e. step **622**).

(34) Referring again to FIG. **8**, at step **640**, in response to determining that the agricultural machine **10** is moving from the field to the road, the controller **86** sends electrical signals to turn on the center valve **74** and turn on the header raise valve **80** to raise the header **22**. As shown at steps **642**, **644**, if the controller **86** determines that the header **22** is positioned above a predetermined or operator-input threshold height (e.g., based on measurements received from the sensors **88**, **92**, **94**), the controller **86** may provide output to the user interface **90** instructing the operator to lock the header **22** in the current position or may, for example, send signals to an actuator (i.e. **42**, **64**, **66**) of the traction assembly **36** causing movement thereof to automatically lock the header **22** in the current position. Thereafter, as shown at step **646**, the controller **86** sends electrical signals to turn

off the center valve **74** and the header raise valve **80**.

(35) Referring still to FIG. **8**, at step **648**, in response to determining that the header **22** is repeatedly moving upward and downward, the controller **86** consults the terrain data if such data is available. At step **650**, the controller **86** sends electrical signals to center valve **74** and at least one of the valves **76**, **78**, **80**, and **82** to move the header **22** upward and downward in response to the ground contours included in the terrain data and the determined position of the header **22** (e.g., based on measurements received from the sensors **88**, **92**, **94**), as described above. Thereafter, as shown at step **652**, the controller **86** sends electrical signals to turn off the center valve **74**, the float valves **76**, **78**, the header raise valve **80**, and the header lower valve **82**.

(36) As shown in FIG. **8**, in each instance of a header height scenario being detected, and subsequent to a header height correction algorithm being executed, the controller **86** executes steps **608** and **610** to return the pressure in the header float valves **64**, **66** to the desired pressure value, as described above.

(37) While this disclosure has been described with respect to at least one embodiment, the present disclosure can be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the disclosure using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this disclosure pertains and which fall within the limits of the appended claims.

Claims

1. An agricultural machine for rapidly moving a header to a harvesting position comprising: a main frame supported above the ground by ground engaging mechanisms; a lift arm pivotably coupled to the main frame; the header which is configured to harvest crop, wherein the header is coupled to and supported above the ground by the lift arm; a lift actuator configured to extend and contract to pivot the lift arm and the header relative to the main frame; a float actuator pivotably coupled to the lift arm and the main frame to support the header above the ground; at least one valve fluidly coupled to the float actuator and configured to move between open and closed positions to change pressure in the float actuator; a controller configured to send electrical signals to the at least one valve to execute a rapid height correction algorithm, in which the controller adjusts at least one of the float actuator and the lift actuator causing a change in height of the header relative to the ground, in response to receiving terrain data including ground contours and receiving correction data associated with at least one of a current height of the header and a current pressure of the float actuator.
2. The agricultural machine of claim 1, further comprising a position sensor operatively coupled to the controller and configured to measure a current height of the header relative to the main frame or relative to the ground; wherein the controller is configured to receive correction data from the position sensor associated with the current height of the header; wherein the controller is configured to compare the current height of the header to the harvesting position of the header, which is a predetermined desired height for the header during a harvesting operation; and wherein the controller is configured to execute the rapid height correction algorithm in response to determining that the current height of the header is beyond a threshold difference from the harvesting position of the header.
3. The agricultural machine of claim 2, wherein in response to determining that the current height of the header is above the harvesting position of the header, the controller sends electrical signals to the least one valve causing the float actuator to extend.
4. The agricultural machine of claim 1, further comprising a pressure sensor operatively coupled to the controller and configured to determine the current pressure of the float actuator; wherein the controller is configured to receive correction data from the pressure sensor indicative of the current

pressure of the float actuator; and wherein the controller is configured to execute the rapid height correction algorithm in response to receiving the correction data from the pressure sensor.
5. The agricultural machine of claim 4, wherein if the correction data received from the pressure sensor is associated with a current height of the header above the harvesting position, then to execute the rapid height correction algorithm the controller sends electrical signals to the least one valve causing the float actuator to extend.
6. The agricultural machine of claim 5, wherein to cause the float actuator to extend, the controller is configured to send a first electrical signal to a first valve causing the first valve to move toward the open position and a second electrical signal to a second valve causing the second valve to move toward the open position.
7. The agricultural machine of claim 4, wherein the correction data includes a signal from the pressure sensor indicative of a decrease in current pressure in the float actuator; and wherein in response to the signal from the pressure sensor indicative of a decrease in current pressure in the float actuator, the controller is configured to send electrical signals to the least one valve causing a further decrease in the current pressure of the float actuator.
8. The agricultural machine of claim 4, wherein the correction data includes a signal from the pressure sensor indicative of an increase in the current pressure of the float actuator; and wherein in response to the signal from the pressure sensor indicative of an increase in the current pressure of the float actuator, the controller is configured to send electrical signals to the least one valve causing a further increase in the current pressure of the float actuator.
9. The agricultural machine of claim 8, wherein the controller is configured to analyze the terrain data including ground contours to determine whether to send electrical signals to the least one valve causing a further increase in the current pressure of the float actuator.
10. The agricultural machine of claim 4, wherein if the correction data received from the pressure sensor is associated with a current height of the header below the harvesting position, then to execute the rapid height correction algorithm the controller sends electrical signals to the at least one valve causing the lift actuator to contract.
11. The agricultural machine of claim 4, wherein if the correction data received from the pressure sensor is associated with a current height of the header below the harvesting position, then to execute the rapid height correction algorithm the controller sends electrical signals to the at least one valve causing the float actuator to contract.
12. The agricultural machine of claim 1, wherein subsequent to execution of the rapid height correction algorithm, the controller is configured to compare a predetermined pressure of the float actuator to the current pressure of the float actuator; and wherein the controller is configured to send additional signals to the at least one valve to adjust the current pressure of the float actuator to match the predetermined pressure of the float actuator.
13. The agricultural machine of claim 1, wherein the correction data comprises global positioning data associated with the agricultural machine performing a 180 degree turn and moving between a field and a road.
14. The agricultural machine of claim 1, wherein the correction data comprises a predetermined operating sequence of the agricultural machine associated with at least one of the agricultural machine performing a 180 degree turn and the agricultural machine moving between a field and a road.
15. An agricultural machine for rapidly moving a header to a harvesting position comprising: a main frame supported above the ground by ground engaging mechanisms; a rockshaft pivotably coupled to the main frame for rotation about a first axis; a lift actuator coupled at a first end to the main frame and at a second end to the rockshaft to facilitate pivoting motion of the rockshaft relative to the main frame; a lift arm pivotably coupled to the main frame for rotation about a second axis parallel to the first axis; a header configured to harvest crop, the header being coupled to and supported above the ground by the lift arm; a float actuator pivotably coupled to the lift arm

for rotation about a third axis, the third axis being parallel to the first axis, wherein the float actuator provides an upward force to the lift arm, thereby cooperating with the lift arm to support the header above the ground; at least one valve fluidly coupled to the float actuator and configured to move between open and closed positions to change the pressure in the float actuator; and a controller configured to send electrical signals to the at least one valve to execute a rapid height correction algorithm, which extends or contracts the float actuator causing a change in height of the header relative to the ground, in response to receiving terrain data including ground contours and receiving correction data associated with at least one of a current height of the header and a current pressure of the float actuator; wherein contraction of the float actuator is configured to occur independent of pivoting motion of the rockshaft relative to the main frame.

16. A method of operating an agricultural machine to rapidly return a header of the agricultural machine to a harvesting position, the method comprising: receiving, from a sensor of the agricultural machine, correction data associated with a current position of the header; comparing the correction data with terrain data including ground contours and the harvesting position of the header, which is a predetermined value associated with a desired height of the header during a harvesting operation; adjusting a float actuator that supports the header above the ground, to move the header to the harvesting position, in response to the determining a difference between the correction data and the harvesting position, wherein the float actuator is coupled between a lift arm that assists in supporting the header above the ground and a main frame of the agricultural machine to which the lift arm is pivotably coupled.

17. The method of claim 16, further comprising: prior to the receiving step, measuring a current pressure of the float actuator via the sensor to obtain the correction data.

18. The method of claim 16, further comprising: prior to the receiving step, measuring the height of the header relative to the ground via the sensor to obtain the correction data.

19. The method of claim 16, further comprising: prior to the receiving step, measuring the height of the header relative to the main frame of the agricultural machine via the sensor to obtain the correction data.

20. The method of claim 16, wherein the adjusting step includes: sending electrical signals to a first valve and a second valve, which are fluidly coupled to the float actuator to cause movement of the first valve and the second valve between open and closed positions.